

Grasp mechanism

Two finger levers on parallel pivots, geared to each other so they move symmetrically. The operator squeezes them by hand.

There is no motor and no CAN on this axis. The levers are purely an input device: a single magnetic rotary encoder on the lever shaft measures how far they have been squeezed, and the console software turns that one number into the patient-side instrument's jaw command. The two wrist axes above it are actuated; this one is not.

Parts

part	file stem	role
Grasp lever 1	grasp_lever_1	finger lever; gear teeth mesh with lever 2
Grasp lever 2	grasp_lever_2	finger lever; mirror of lever 1
Grasp mount plate	grasp_mount_plate	carries the lever pair
Grasp pivot pins	grasp_pin_1 , grasp_pin_2	lever pivots, 2 mm dia x 10.5 mm
Grasp sensor base	grasp_sensor_base	levers pivot in its centre holes; the encoder mounts on top
Grasp cover	grasp_cover	lever housing
MCU base	mcu_base	carries the XIAO board

CAD for the two levers is in `../CAD/grasp_module/` ([grasp_lever_1](#), [grasp_lever_2](#)).

[grasp_sensor_carrier](#) in the same folder holds the sensing parts as three solids: the magnet (5 mm dia x 2 mm), the sensor IC and its PCB. The other printed parts of the module are released as STL only.

Kinematics

joint	type	parent	axis	actuated
gear_1_joint	revolute	link_2dof	-X in the link_2dof frame	no — hand-driven
gear_2_joint	revolute	link_2dof	-X in the link_2dof frame	no — hand-driven

The gear teeth constrain the pair, so one angle describes both. The software publishes one lever angle and mirrors it onto the other ([gear_2](#) = -[gear_1](#)).

No joint limits are set in the CAD; the travel is bounded by the parts themselves. The URDF limits each lever joint to $\pm 30^\circ$ (0.5236 rad). Measured at the encoder, the full travel has come out between about **20 and 36 degrees** depending on the unit, so treat the range as something to measure per unit, not a design constant.

The controller loads the URDF with both lever joints **locked at neutral**. That is not bookkeeping: the dynamics library counts every unlocked joint, so leaving the levers free changes the model's DOF count and breaks code that assumes the arm's. Any joint added to the URDF beyond the wrist has to be locked the same way.

Sensing

encoder	AS5600-class magnetic rotary, 12-bit, 4096 counts per revolution
mounting	on the lever shaft, carried by grasp_sensor_base
front end	Seeed XIAO nRF52840 Sense on mcu_base , USB 2886:8045
stream	about 167 Hz, CSV over USB CDC; the angle field is already in degrees
normalised output	0.0 = fully open, 1.0 = fully closed

Two things about this sensor decide whether the grasp feels right:

The magnet wraps at 0/360. The host driver unwraps the reading into a cumulative angle, but if the lever's working range straddles the wrap point, a squeeze right at the boundary makes the jaw jump open. Mount the magnet so the whole travel sits away from the seam.

The endpoints are per-unit, not per-design. Fully-open and fully-closed angles come from a calibration run on each assembled gripper. So do the sign and the scale, because they depend on how the sensor is wired and which way the magnet faces. None of these belong in the CAD or the URDF.

Build checks

- The two levers move symmetrically. A half-tooth offset at assembly is a permanent bias the software cannot see.
- The encoder reading sweeps smoothly across the full lever travel. A reading that never moves by more than one count means the magnet is too far away or not centred.
- The travel does not cross the 0/360 seam.